Passive Risk Detection Report

devdaytest1 . Harness CD . Cluster: validation-cluster . Namespace: default

Scan Identity: test-scan-1778747446 . Scanned 2026-05-15 07:06 UTC

HARNESS RESILIENCE TESTING -- EXPERIMENTAL | Confidential

1 . Executive Summary

58/1000
PIPELINE RISK SCORE

5
HIGH RISKS

2
MEDIUM RISKS

0
LOW RISKS

7

This report presents the results of a passive resilience risk scan performed by the Harness Resilience Testing engine against Harness CD pipeline for project devdaytest1. The scan targeted cluster validation-cluster in namespace default. No code changes, instrumentation, or manual configuration were required -- the engine scanned Harness CD metadata, Kubernetes manifests, and deployment configuration only.

The scan identified 5 HIGH-severity risks and 2 MEDIUM-severity risks across 1 service(s). All 7 risks are currently in Open status with no chaos or load tests covering them -- a critical coverage gap.

Metric	Value	Detail
Services Scanned	1	Deployment in namespace: default
Total Risks Detected	7	High: 5, Medium: 2, Low: 0
Services Affected	1	nginx
Chaos Test Coverage	0/1	100% uncovered -- no resilience validation
Open Risks	7	All risks remain unresolved
Scan Type	kubernetes-security	Passive metadata-only analysis

2 . Services Found in Pipeline

All services in the target namespace were passively scanned -- no code changes, instrumentation, or manual configuration required.

Service	Total Risks	High	Medium	Low	RRS
nginx	7	**5**	**2**	**--**	**58**
TOTAL (1 svc)	**7**	**5**	**2**	**0**	**58**

** RRS = HIGH x 10 + MEDIUM x 4 + LOW x 1. Critical (>=30): immediate action required. High (15-29): near-term attention. Medium (<15): best-practice deviation.*

3 . Risks Found

7 risks detected across 1 service(s) -- 5 HIGH . 2 MEDIUM . 0 LOW. Each risk listed with description and affected services.

HIGH

k8s-missing-resource-limits

The nginx container has no CPU or memory limits. It can consume all available node resources, starving sibling pods.

Affects: Deployment/nginx in namespace hce

Impact: Unbounded resource consumption causes noisy-neighbour failures on the node. Memory hog can trigger kernel OOM kills on sibling pods.

kubernetes reliability nginx

HIGH

k8s-single-replica

The nginx Deployment runs with replicas: 1, making it a single point of failure. A pod-delete fault causes 100% service unavailability until the pod restarts.

Affects: Deployment/nginx in namespace hce

Impact: Complete service downtime during any pod disruption, node maintenance, or OOM kill. No redundancy to absorb pod-level failures.

kubernetes availability nginx

MEDIUM

k8s-missing-resource-requests

The nginx container has no CPU or memory requests. It is assigned BestEffort QoS and will be the first pod evicted under any node pressure.

Affects: Deployment/nginx in namespace hce

Impact: BestEffort QoS means nginx is evicted before any other pod on the node when memory or CPU pressure occurs -- even if nginx itself is not causing the pr...

kubernetes reliability nginx

HIGH

k8s-missing-probes

The nginx container has no livenessProbe or readinessProbe. Kubernetes routes traffic to unhealthy pods and never restarts stuck ones.

Affects: Deployment/nginx in namespace hce

Impact: Traffic is sent to pods that are restarting or deadlocked. Without livenessProbe, a hung nginx process is never restarted automatically.

kubernetes availability nginx

HIGH

k8s-missing-network-policy

The nginx Deployment has no NetworkPolicy. All ingress and egress traffic is unrestricted, enabling lateral movement from any compromised pod in the cluster to reach nginx directly.

Affects: Deployment/nginx in namespace hce

Impact: Any pod in any namespace can send traffic to nginx without restriction. A compromised workload can exfiltrate data, probe nginx endpoints, or use it a...

kubernetes security network nginx

MEDIUM

k8s-missing-pod-anti-affinity

The nginx Deployment has no podAntiAffinity or topologySpreadConstraints. All replicas may be scheduled on the same node, turning the multi-replica deployment into a node-level single point of failure.

Affects: Deployment/nginx in namespace hce

Impact: Despite having replicas configured, all pods could land on the same node. A single node-drain or node-restart takes down every replica simultaneously ...

kubernetes availability nginx

HIGH

k8s-latest-image-tag

The nginx container uses image 'nginx' with no tag. A pod restart may pull a different version from the registry, causing silent version drift between replicas.

Affects: Deployment/nginx in namespace hce

Impact: Non-deterministic deployments -- two pods in the same Deployment can silently run different image versions after independent restarts.

kubernetes reliability nginx

4 . Severity Distribution & Risk Category Breakdown

The 7 total risks span multiple categories affecting the target deployment.

Severity Level	Count	% of Total	Definition
HIGH (Critical)	**5**	71%	High-confidence failure mode with direct outage potential
MEDIUM (Warning)	**2**	29%	Elevated risk requiring near-term attention
LOW (Info)	**0**	0%	Low-severity observation or best-practice deviation
TOTAL	**7**	**100%**	**Across all services in scan scope**

Risk Count by Category

Category	Count	Risks Included
Availability	**3**	Single-replica, missing probes, pod anti-affinity
Reliability	**3**	Missing resource limits/requests, latest image tag
Security	**1**	Missing network policy, lateral movement risk

5 . Recommendations & Next Steps

Prioritised actions to reduce the pipeline risk score from 58 to below the recommended gate threshold of 100 within the next two sprint cycles.

Priority	Service	Action	Experiment Type
P1	**nginx**	Add CPU/memory limits	Memory Stress
P1	**nginx**	Increase replicas + add PDB	Pod Delete
P2	**nginx**	Add resource requests for QoS	Pod Delete (pressure)
P1	**nginx**	Add liveness/readiness probes	Pod Delete + traffic
P1	**nginx**	Create NetworkPolicy	Network Partition
P2	**nginx**	Add podAntiAffinity	Node Drain
P1	**nginx**	Pin image to version/SHA	Rolling Update

Recommended Immediate Action:

Run a pod-delete chaos experiment against nginx -- the deployment is single-replica with no health probes and zero resilience test coverage. This will immediately confirm or disprove complete service outage on pod eviction.

Auto-parameterised experiment config:

target namespace = default, blast radius = 100%, steady-state hypothesis = HTTP 200 on health endpoint at p99 < 500ms, duration = 5 min, rollback on error rate > 2x baseline.

6 . Test Coverage Gap Analysis

Zero coverage means a service's resilience posture is entirely unverified -- all passive risks remain unconfirmed and no steady-state baseline has been established.

Service	Chaos Tests	Load Tests	Coverage %	Notes
nginx	None -- Gap	None -- Gap	**0%**	Critical: no tests of any kind
SUMMARY	**0/1 covered**	**0/1 covered**	**Avg 0%**	**Immediate action required**

Services with no tests of any kind: nginx -- 1 of 1 services.

. Appendix -- About This Report & Glossary

This report was generated by the Harness Resilience Testing PRD-Report API (experimental). It is produced automatically at the end of each CD pipeline execution without requiring any code changes, instrumentation, or manual scan initiation. All findings are derived from static metadata analysis only -- no live traffic was affected and no faults were injected to produce this report.

Risk taxonomy follows the classification defined in Harness Resilience Testing PRD Section 5A (v0.1, May 2026). Monetary estimates are illustrative and not contractual.

Term	Definition
RRS	Resilience Risk Score -- composite: HIGH x 10 + MEDIUM x 4 + LOW x 1
PDB	PodDisruptionBudget -- K8s policy limiting simultaneous pod evictions
HPA	HorizontalPodAutoscaler -- auto-scales pod count based on load metrics
MTTR	Mean Time To Recovery -- average time to restore service after incident
PRD-Report	Passive Risk Detection Report -- API-driven risk scan anchored to CD pipeline
HIGH	High-confidence failure mode with direct outage potential
MEDIUM	Elevated risk requiring near-term attention
LOW	Low-severity observation or best-practice deviation